

Without rounds, a swarm holds its proof less firmly

Simulation 2 (asynchronous activation), setup `task003-symmetric`: first results

LLM-free simulator, runs of 9 October 2026

Summary. The 24 agents together hold a proof of allocation A0 (the truth). Simulation 2 removes the day/night rounds of Simulation 1: agents and the controller act at random times, remembered facts fade continuously, and the shared board fades too. When agents forget, **the swarm holds its own proof less firmly** than in Simulation 1, and **the false controller is weaker** (gain 0.13 against 0.22). How fast the controller acts matters little for where the swarm ends up, but **fast controllers overshoot and then relax**. Turning the vote gate off doubles the truth controller’s gain at six times the messages. As in Simulation 1, help with the truth **replaces the agents’ own proofs**. Effects persist for ten time units after the controller stops. Main hypotheses were confirmed on a second, independent set of random draws.

1 Setup

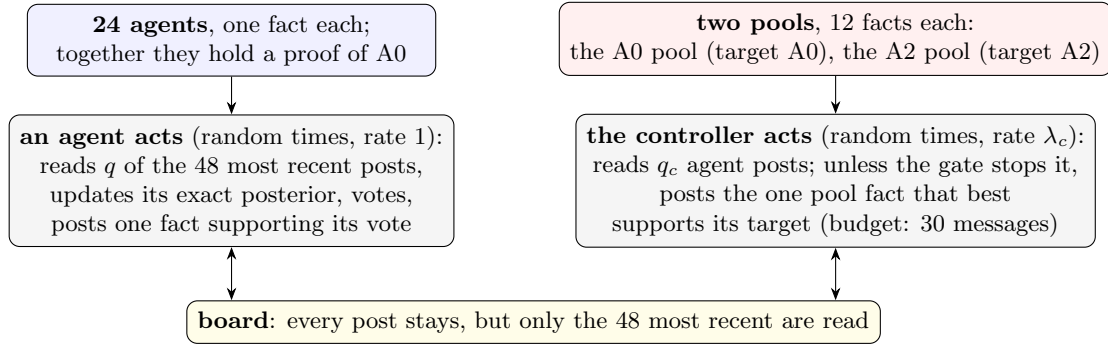


Diagram 1. Simulation 2. There are no rounds. Each remembered fact is forgotten after a random lifetime (on average a fraction ρ survives one time unit) and comes back if read again. The controller acts for $0 \leq t < 30$; agents continue alone until $t = 40$.

Agents are exact Bayesian reasoners. An agent’s vote is drawn from its posterior (probability matching). The *vote gate* keeps the controller silent when at least 75% of the posts it read come from agents who voted for its target; the *proof stop* keeps it silent when what it read already proves its target. The grid: posts read $q \in \{3, 6, 12\}$, $q_c \in \{3, 6, 12, 24\}$, memory $\rho \in \{0.75, 1\}$, controller rate $\lambda_c \in \{0.5, 1, 2, 4, 8\}$ actions per time unit, gate on/off, proof stop on/off; 1,000 episodes per setting. Control is measured by the *gain*

$$G(t) = \frac{1}{E} \sum_{e=1}^E \left[m_{\text{target}}(t)^{\text{ctrl}}(e) - m_{\text{target}}(t)^{\text{silent}}(e) \right], \quad (1)$$

where $m_k(t)$ is the agents’ mean belief in allocation k and each episode e uses the same random draws with and without the controller. Unless stated otherwise: $q = 6$, $q_c = 12$, $\rho = 0.75$, gate on, proof stop on.

2 Without a controller: the proof is held less firmly

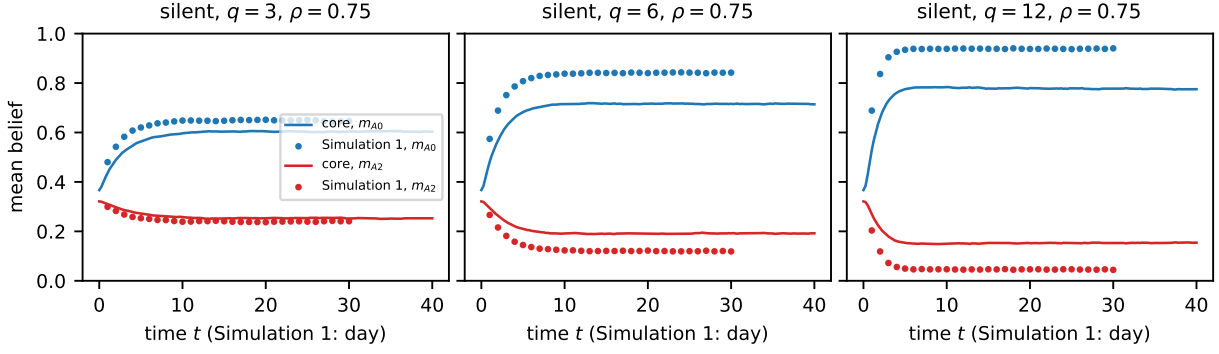


Figure 1: Silent swarm, $\rho = 0.75$: mean belief in A0 (blue) and A2 (red) over time. Dots: Simulation 1, end of each day. Lines: Simulation 2.

With forgetting, the silent swarm settles at a lower belief in the truth than in Simulation 1: $m_{A0}(30) = 0.60/0.72/0.78$ against $0.65/0.84/0.94$ for $q = 3/6/12$ (Figure 1); confirmed with fresh random draws (-0.12 at $q = 6$). In Simulation 1 memories change only at dawn, and every agent acts right after a full day of new posts. Here facts expire at random moments between an agent’s actions, so fewer facts are held together at any time and complete proofs are rarer. With full memory ($\rho = 1$) both simulations end with every agent proving A0, and no controller can change that: all gains at $\rho = 1$ are zero.

3 Control

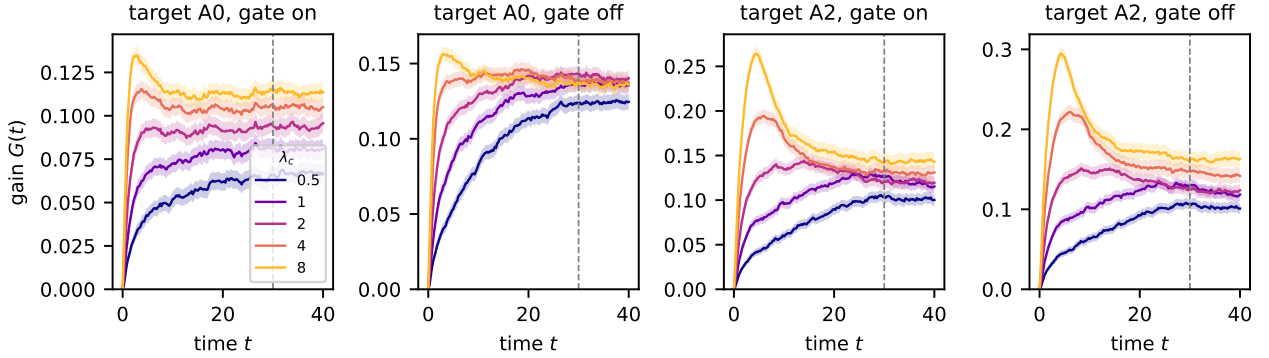


Figure 2: Gain over time, Eq. (1), by controller rate λ_c ; the controller stops at $t = 30$ (dashed). Bands: 95% intervals.

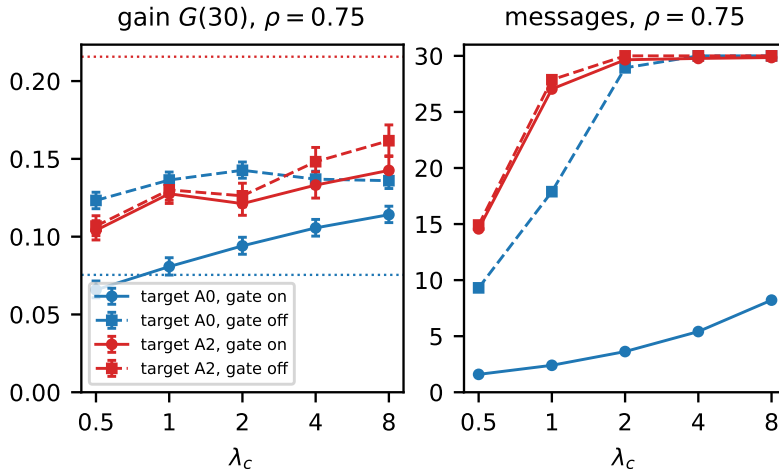


Figure 3: Left: gain at $t = 30$ against the controller rate (target A0 blue, A2 red; gate on solid, off dashed; dotted: Simulation 1 with one fact per night). Right: messages sent per episode (budget 30).

- **The false controller is weaker than in Simulation 1:** $G(30) = 0.13$ against 0.22 (one fact per night in Simulation 1); confirmed with fresh draws (-0.09). The truth controller is unchanged (0.08 in both).
- **The rate changes the path more than the end point** (Figures 2, 3). From $\lambda_c = 0.5$ to 8 the gain at $t = 30$ rises from 0.10 to 0.14 (target A2) and from 0.07 to 0.11 (target A0). A fast controller spends its budget within a few time units, its gain peaks early and then relaxes: target A2 at $\lambda_c = 8$ peaks at 0.26 ($t \approx 4.5$) and falls to 0.14.
- **The effect persists:** ten time units after the controller stops, the gain has changed by at most 0.021 in every setting, less than one agent's worth of belief ($1/24 \approx 0.042$); confirmed with fresh draws.

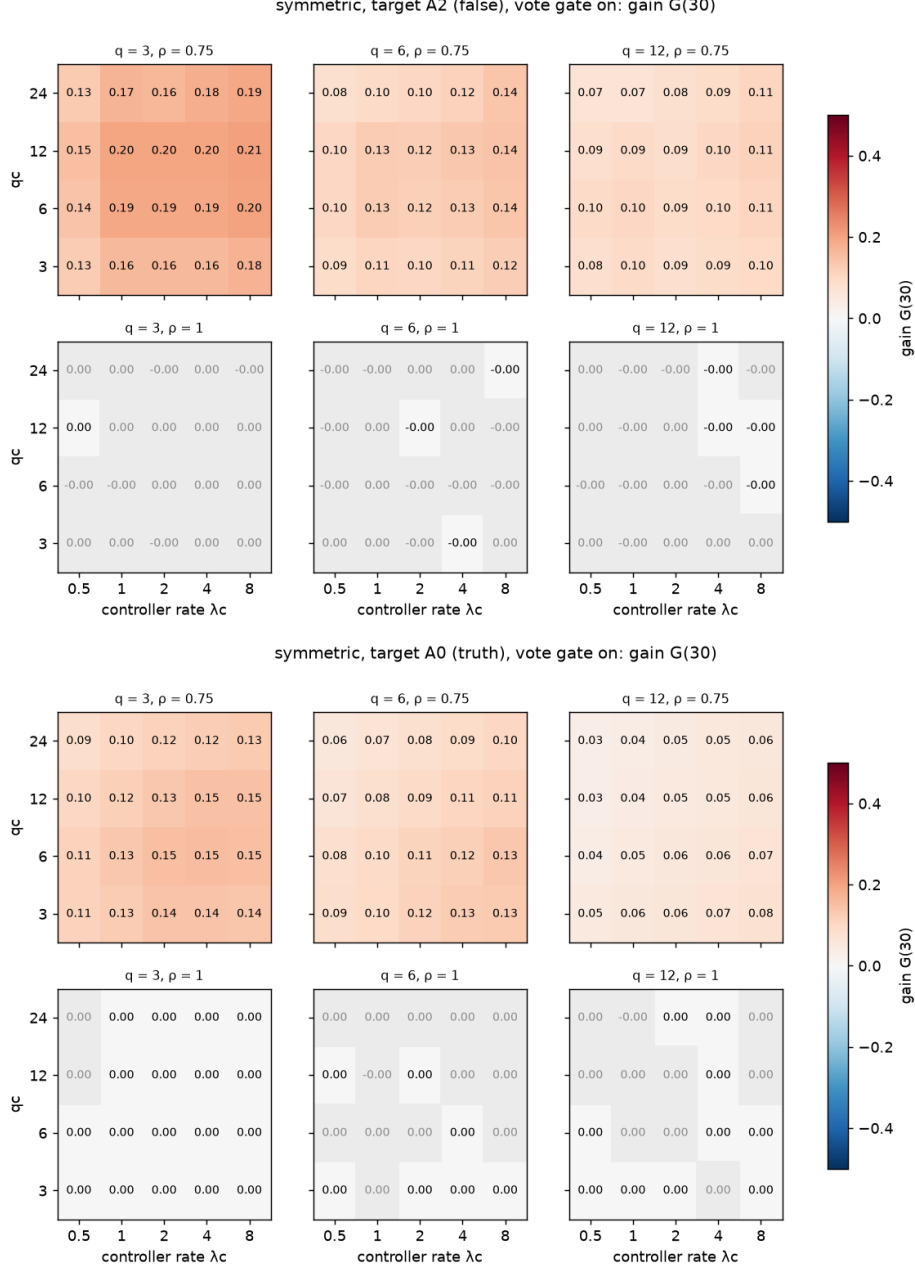


Figure 4: Gain $G(30)$ over controller rate (x) and posts read by the controller q_c (y), for every q (columns) and ρ (rows); gate on. Top: target A2; bottom: target A0. Grey numbers: 95% interval includes 0.

4 The vote gate and the proof stop

The false controller is almost never stopped by the gate (the A2 share rarely reaches 75%), so turning the gate off changes nothing for it. The truth controller often is: with the gate off it sends about six times as many messages (9.3 against 1.6 at $\lambda_c = 0.5$) and its gain roughly doubles ($0.066 \rightarrow 0.123$; $+0.055$ at $\lambda_c = 1$, confirmed

with fresh draws). The proof stop has no effect larger than one agent’s worth of belief ($1/24$) in any setting, and turning it off costs up to 25 extra messages per episode.

5 Help with the truth replaces the agents’ own proofs

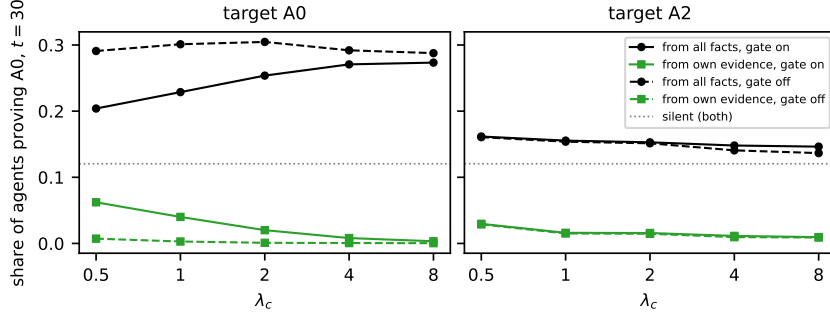


Figure 5: Share of agents whose memory proves A0 at $t = 30$: from all their facts (black) and from the agents’ original facts only (green); dotted: silent swarm.

Without a controller, 12% of agents prove A0 at $t = 30$, all from the agents’ own facts. The truth controller raises the share proving A0 to 20–30%, but the share proving it *from the agents’ own facts* falls to 6% (slow, gated controller) and to almost 0 when it posts more (Figure 5). Agents posting a controller fact do not post their own, so their own evidence circulates less. The false controller, whose facts are also true, raises proofs of A0 too (to 14–16%) while own-evidence proofs fall to 1–3%. Simulation 1 showed the same substitution.

6 The controller overestimates its support

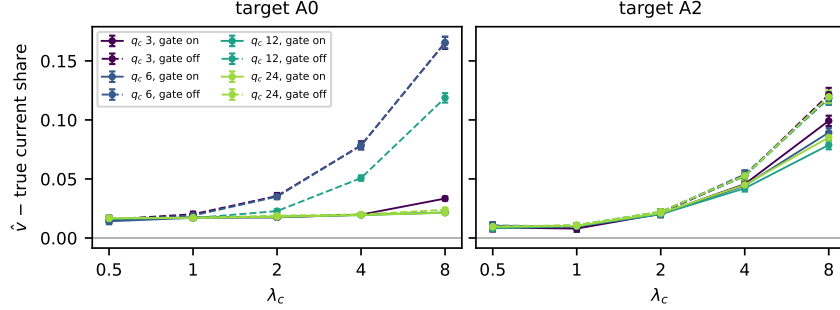


Figure 6: Error of the controller’s estimate $\hat{v}$ (share of posts read whose author voted for the target) against the true current share of all 24 agents’ last votes; solid gate on, dashed off.

Unlike in Simulation 1, $\hat{v}$ is biased upward, and **the faster the controller, the larger the bias**: about $+0.014$ at $\lambda_c \leq 1$, rising to $+0.05$ (gate on) and $+0.14$ (gate off) at $\lambda_c = 8$. A likely reason, not yet tested: the posts it reads come from agents who acted recently, often just after reading the controller’s facts, while the true share also counts older votes. More posts read (q_c) reduce the error’s spread but not the bias.

Caveats. One world (task_003), 24 agents, exact-Bayesian agents, probability-matching votes; a controller without memory that posts one fact at a time; a budget of 30 messages. Means over 1,000 episodes per setting. Full analysis, code and tables: analysis/task003.llm_free/simulation_2/analysis/ (report of 9 October, reviewer guide).